

Exp No.: 4 Querying Ledger Data with Hyperledger Fabric

Aim

To query and retrieve ledger data from a Hyperledger Fabric blockchain network using JavaScript chaincode, and verify asset information stored in the world state database.

Software Requirements

- Kali Linux / Ubuntu
- Docker
- Docker Compose
- Node.js (Version 18 or later)
- npm
- Git
- Hyperledger Fabric Samples
- Hyperledger Fabric Binaries
- Hyperledger Fabric Docker Images
- Visual Studio Code (Optional)

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB Free Disk Space
- Internet Connection

Procedure

Step 1: Navigate to the Fabric Test Network

Open the terminal and navigate to the Hyperledger Fabric test network.

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Hyperledger Fabric Network

Start the Fabric network and create the application channel.

```
./network.sh up createChannel -ca
```

Step 3: Deploy the Asset Transfer Chaincode

Deploy the JavaScript Asset Transfer chaincode.

```
./network.sh deployCC \  
-ccl javascript \  
-ccn basic \  
-ccp ../asset-transfer-basic/chaincode-javascript
```

Step 4: Initialize the Ledger

Load the ledger with sample assets.

```
peer chaincode invoke ... -c '{"function":"InitLedger","Args":[]}'
```

```
kali@kali: ~/fabric-dev/fabric-samples/test-network

Session Actions Edit View Help

basic_1.0:1d4d445573112813889f87c307bc2b5f6e664c87b24c035bee15cbcc7f59b629
Chaincode is installed on peer0.org2
++ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
++ sed -n '/Version:/{s/.*Sequence: //; s/, Endorsement Plugin:.*$/; p;}'
Error: query failed with status: 404 - namespace basic is not defined
+ COMMITTED_CC_SEQUENCE=
+ res=0
Using organization 1
+ peer lifecycle chaincode queryinstalled --output json
+ grep '^basic_1.0:1d4d445573112813889f87c307bc2b5f6e664c87b24c035bee15cbcc7f59b629$'
+ jq -r 'try (.installed_chaincodes[].package_id)'
+ res=0
basic_1.0:1d4d445573112813889f87c307bc2b5f6e664c87b24c035bee15cbcc7f59b629
Query installed successful on peer0.org1 on channel
Using organization 1
+ peer lifecycle chaincode approveformyorg -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile /home/kali/fabric-dev/fabric-samples/test-network/organizations/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem --channelID mychannel --name basic --version 1.0 --package-id basic_1.0:1d4d445573112813889f87c307bc2b5f6e664c87b24c035bee15cbcc7f59b629 --sequence 1
+ res=0
2026-09-08 11:44:48.838 EDT 0001 INFO [chaincodeCmd] ClientWait → txid [92d53bd8621cf4c7c08a893cb6dcf19ebb66b1660ee07fdfbe54a09feb4b64ff] committed with status (VALID) at localhost:7051
Chaincode definition approved on peer0.org1 on channel 'mychannel'
Using organization 1
Checking the commit readiness of the chaincode definition on peer0.org1 on channel 'mychannel'...
Attempting to check the commit readiness of the chaincode definition on peer0.org1, Retry after 3 seconds.
+ peer lifecycle chaincode checkcommitreadiness --channelID mychannel --name basic --version 1.0 --sequence 1 --output json
+ res=0
{
  "approvals": {
    "Org1MSP": true,
    "Org2MSP": false
  }
}
Checking the commit readiness of the chaincode definition successful on peer0.org1 on channel 'mychannel'
Using organization 2
Checking the commit readiness of the chaincode definition on peer0.org2 on channel 'mychannel'...
Attempting to check the commit readiness of the chaincode definition on peer0.org2, Retry after 3 seconds.
+ peer lifecycle chaincode checkcommitreadiness --channelID mychannel --name basic --version 1.0 --sequence 1 --output json
+ res=0
{
  "approvals": {
    "Org1MSP": true,
    "Org2MSP": false
  }
}
Checking the commit readiness of the chaincode definition successful on peer0.org2 on channel 'mychannel'
```

Figure 1: Chaincode installation, approval, and commit readiness check across both organizations

Step 5: Query All Assets

Retrieve all asset records stored in the blockchain ledger.

```
peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
```

Step 6: Query a Specific Asset

Retrieve the details of asset1.

```
peer chaincode query -C mychannel -n basic -c  
'{"Args":["ReadAsset","asset1"]}'
```

Step 7: Create a New Asset

Create a new asset (asset11) on the ledger.

```
peer chaincode invoke ... -c  
'{"function":"CreateAsset","Args":["asset11","purple","12","Arun","900"]}'
```

Step 8: Query the Newly Created Asset

Read back asset11 to verify it was created.

```
peer chaincode query -C mychannel -n basic -c  
'{"Args":["ReadAsset","asset11"]}'
```

Step 9: Stop the Fabric Network

Shut down all containers, volumes, and chaincode images.

```
./network.sh down
```

```

kali@kali: ~/fabric-dev/fabric-samples/test-network
Session Actions Edit View Help

(kali@kali)~/fabric-dev/fabric-samples/test-network
$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset1"]}'
{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"}

(kali@kali)~/fabric-dev/fabric-samples/test-network
$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile "$ORDERER_CA" -C mychannel -n basic -c '{"function":"CreateAsset","Args":["asset11","purple","12","Arun","900"]}' --peerAddresses localhost:7051 --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org1.example.com/tlsca/tlsca.org1.example.com-cert.pem" --peerAddresses localhost:9051 --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem"
2026-09-08 11:48:29.314 EDT 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status: 200 payload: '{"ID":"asset11","Color":"purple","Size":12,"Owner":"Arun","AppraisedValue":900}'

(kali@kali)~/fabric-dev/fabric-samples/test-network
$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset11"]}'
{"AppraisedValue":900,"Color":"purple","ID":"asset11","Owner":"Arun","Size":12}

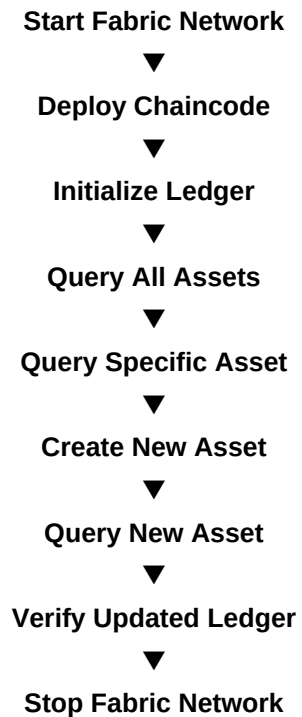
(kali@kali)~/fabric-dev/fabric-samples/test-network
$ ./network.sh down
Using docker and docker-compose
Stopping network
[+] Running 10/10
✔ Container peer0.org1.example.com Removed 0.9s
✔ Container ca_orderer Removed 10.5s
✔ Container peer0.org2.example.com Removed 0.8s
✔ Container orderer.example.com Removed 0.4s
✔ Container ca_org2 Removed 10.4s
✔ Container ca_org1 Removed 10.6s
✔ Volume compose_peer0.org2.example.com Removed 0.0s
✔ Volume compose_orderer.example.com Removed 0.0s
✔ Volume compose_peer0.org1.example.com Removed 0.0s
✔ Network fabric_test Removed 0.2s
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
Removing generated chaincode docker images
Untagged: dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5f5bee15cbcc7f59b629-c68f8ecc2a0ff0cc1497615f5e33a8bfa908e0fd3555f864cfa1398b52d341cb:latest
Deleted: sha256:5dd0a7e81dd28890b0c88f999be0368d550ff83b32c34759c8275d41e0b60b6b
Untagged: dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5f5bee15cbcc7f59b629-d3cf9b1f4b9747cc8433ac0e74bfaeade1cf0b97f5c1163835f604aa885d60dc:latest
Deleted: sha256:11cb964a311d6cb765a1c704664639698bb5a77dd17537e56efda70949661a0c

(kali@kali)~/fabric-dev/fabric-samples/test-network
$

```

Figure 2: Querying asset1, creating asset11, verifying it, and stopping the network

Flow of Ledger Query Process



Result

The ledger data stored in the Hyperledger Fabric blockchain network was successfully queried using JavaScript chaincode. Asset records were retrieved from the world state database, specific asset details were verified, and newly created assets were successfully queried, demonstrating efficient data retrieval and verification in a permissioned blockchain environment.